

# RUTHERFORD CALAWAGAN

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## EDUCATION

### University of British Columbia

Sep 2023 – Dec 2027 (Expected)

Bachelor of Computer Science | 3.8 GPA, Dean's List

Vancouver, BC

- **Relevant Coursework:** Algorithms & Data Structures, Computer Systems, Software Construction, Models of Computation

## WORK EXPERIENCE

### Trulioo | Software Engineer Intern

May 2026 – Present

React, TypeScript, C#, .NET

Vancouver, BC

- Reduced overhead in an internal encryption library, cutting peak memory usage by **80.4%** and encryption/decryption time by **50%**, by precomputing message size and replacing intermediate streams with a single preallocated buffer.
- Increased observability across asynchronous processing workflows, instrumenting **5 workflows and 52 stages**, by implementing step-level execution timing with `AsyncLocal` and an accessor abstraction for context propagation.
- Streamlined production logging across 9 services, reducing disk usage by **80%**, by standardizing NLog retention and correcting archive rotation across date boundaries.
- Automated .NET 8-to-10 modernization across 7 services, updating **20+ dependencies** and remediating **30+ vulnerabilities**, by building a reusable coding-agent SKILL that generated review-ready merge requests.

### Procurify | Software Engineer Intern

Sep 2025 – Apr 2026

React, TypeScript, Python, Django, FastAPI

Vancouver, BC

- Accelerated export-service performance, cutting database calls by **99% (3,200+ to 14)** and request latency by **25%**, by eliminating **N+1 SQL queries** and introducing batched data retrieval.
- Reduced memory pressure in the audit-log version-history pipeline, lowering peak usage by **84% (1.55 GB to 253 MB)**, by re-architecting version-history construction around **ID-based batching**.
- Strengthened concurrency correctness in integration-status updates, eliminating a race condition under concurrent workflows, by introducing **row-level locking** with PonyORM's `get_for_update()`.
- Improved user-data visibility for account management and troubleshooting by shipping customer-facing React features and supporting APIs in collaboration with senior engineers, product, and design.

### University of British Columbia | Software Development - Research Assistant

Apr 2025 – Jul 2025

AI in Finance

Vancouver, BC

- Accelerated 60-day ML preprocessing by **~25%** while replicating a published pipeline across 5- and 60-day horizons, by reusing parsed market data and normalized price arrays.

## PROJECTS

### RepPilot [Live App] | Next.js, TypeScript, PostgreSQL, Supabase, OpenAI API, Vitest

Jun 2026 – Aug 2026

- Built a mobile-first workout platform with Next.js and TypeScript, designing a **normalized PostgreSQL schema** for templates, sessions, exercises, and set-level performance history.
- Engineered a **deterministic progression engine** with pain overrides, stall detection, and reproducible recommendations; used OpenAI to explain decisions while enforcing per-user isolation through Supabase row-level security.

### FoodRadar [GitHub] | JavaScript, React, Node.js/Express, MongoDB, Google Cloud Vision

Apr 2025

- Expanded a hackathon-built MERN inventory application with Google Cloud Vision scanning and secure JWT and OAuth 2.0 authentication.

## LEADERSHIP & ADDITIONAL EXPERIENCE

### UBC BCSSA | Professional Committee

Jan 2026 – Present

- Coordinate career-development events for BCS students, managing speaker outreach, logistics, and promotion in partnership with the VP Professional.

### Unilever | Future Leaders Program – Supply Chain

Aug 2021 – Jul 2024

- Completed a three-year management rotation across Integrated Planning, Manufacturing, Customer Service, and Logistics, delivering cross-functional projects including a Python and Power BI tool that increased service level from **78% to 90%**.

## TECHNICAL SKILLS

**Languages:** C#, Python, TypeScript/JavaScript, Java, C++, SQL, HTML/CSS

**Frameworks:** .NET, React, Next.js, Django, FastAPI, Node.js/Express

**Cloud & Tools:** AWS (ECS, S3, RDS, SQS, Kinesis Data Streams), Docker, Bash, Git, GitHub, GitLab, CircleCI

**Databases & Observability:** PostgreSQL, MySQL, MongoDB, Grafana, Kibana, Loki